

MATH3303 – Lecture 32

Uniqueness in UFDs; Irreducible Polynomials

Review Notes

Reference: Lee, Abstract Algebra, Sections 10.4–11.1

Overview

This lecture

- Completes the proof that every PID is a UFD by establishing uniqueness of factorisation (Claim 2).
- Records two important corollary remarks: in any UFD, irreducible $\iff$ prime; and R UFD implies $R[x]$ UFD.
- Opens a new chapter on *irreducible polynomials* over a field, starting from the Remainder and Factor Theorems.

1 Every PID is a UFD: Uniqueness of Factorisation

Recall (Lecture 31) the theorem: every PID is a UFD. Claim 1 (existence of a factorisation into irreducibles) was proved last lecture via the ACC. We now finish the proof.

Proof of “every PID is a UFD”, Claim 2 (uniqueness). Let R be a PID. We must show that the factorisation in Claim 1 is unique in the sense of the UFD definition.

Suppose

$$p_1 \cdots p_k = q_1 \cdots q_\ell,$$

where each p_i and q_j is irreducible in R . WLOG assume $k \leq \ell$.

Step 1: p_1 divides some q_i . We have $p_1 \mid q_1 \cdots q_\ell$. Since R is a PID, irreducibles are prime (Lecture 31). So $p_1 \mid q_i$ for some i . WLOG (relabelling), assume $p_1 \mid q_1$. Write

$$q_1 = p_1 u_1 \quad \text{for some } u_1 \in R.$$

Since q_1 is irreducible and p_1 is not a unit, u_1 must be a unit.

Step 2: cancel and recurse. Substituting back:

$$p_1 \cdots p_k = u_1 p_1 q_2 \cdots q_\ell.$$

Cancel p_1 (allowed: R is an integral domain, $p_1 \neq 0$):

$$p_2 \cdots p_k = u_1 q_2 \cdots q_\ell.$$

Now p_2 divides $u_1 q_2 \cdots q_\ell$. Since p_2 is prime, it divides at least one factor.

Sub-step: p_2 cannot divide u_1 . If $u_1 = u' u_1'$ with $p_2 \mid u_1$, then since u_1 is a unit, write $1 = u_1 u_1^{-1}$, so any divisor of u_1 is a unit. (Explicitly: if $u = vw$ then $1 = uu^{-1} = vwu^{-1}$, so v is a unit.) But p_2 is irreducible and not a unit. Contradiction.

Hence $p_2 \mid q_i$ for some $i \geq 2$. Relabel so $p_2 \mid q_2$, then write $q_2 = u_2 p_2$ for some unit u_2 , and cancel.

Step 3: repeat. Continuing for all p_i , $2 \leq i \leq k$, we eventually obtain

$$1 = u_1 u_2 \cdots u_k q_{k+1} \cdots q_\ell.$$

If $k < \ell$, then $q_{k+1}, \dots, q_\ell$ must all be units. But each q_j is irreducible, hence not a unit. Contradiction.

Therefore $k = \ell$, and after permutation each p_i and q_i are associates. □

This completes the proof of the theorem: **every PID is a UFD.**

Remark 1.1: Irreducible $\iff$ prime in a UFD (from lecture)

In a UFD, an element is irreducible if and only if it is prime.

($\Leftarrow$) Prime always implies irreducible in any integral domain (Lecture 30).

($\Rightarrow$) Suppose p is irreducible in a UFD R , and $p \mid ab$ with $a, b \in R$. Write $ab = pc$ for some $c \in R$. Factor each of a, b, c into irreducibles:

$$a = p_1 \cdots p_r, \quad b = q_1 \cdots q_s, \quad c = r_1 \cdots r_t.$$

Then

$$p_1 \cdots p_r q_1 \cdots q_s = p r_1 \cdots r_t.$$

By unique factorisation, p must be associate to one of the irreducibles on the left.

- If p and some p_i are associates, then $p = up_i$ for some unit u , so $p_i \mid a$ implies $p \mid a$.
- If p and some q_j are associates, similarly $p \mid b$.

Either way, $p \mid a$ or $p \mid b$. Hence p is prime. $\square$

Remark 1.2: If R is a UFD, then $R[x]$ is a UFD (from lecture)

If R is a UFD, then so is $R[x]$. For example:

- $\mathbb{Z}$ a UFD $\Rightarrow \mathbb{Z}[x]$ a UFD.
- K a field (hence a UFD) $\Rightarrow K[x]$ a UFD $\Rightarrow K[x, y] = (K[x])[y]$ a UFD.

The proof is rather tricky and is deferred. The key tool is to pass to the *fraction field*:

$$R[x] \rightsquigarrow K[x], \quad K := \text{Frac}(R).$$

Here $K = \text{Frac}(R)$ is the *fraction field* of R , constructed as the set of formal symbols $\{a/b\}$ with $b \neq 0$, modulo the equivalence $a/b \sim c/d \iff ad = bc$. For example, $\text{Frac}(\mathbb{Z}) = \mathbb{Q}$, so $\mathbb{Z}[x] \rightsquigarrow \mathbb{Q}[x]$.

2 Irreducible Polynomials over a Field

We now begin a new topic: irreducibility of polynomials in $K[x]$, where K is a field.

Setup. Let K be a field. Every degree-1 polynomial in $K[x]$ is irreducible. In general, however, irreducibility of a polynomial depends on the *ground field*.

Example 2.1: $x^2 - 2$ depends on the ground field (from lecture; cf. Lee, Example 11.1)

Let $f(x) = x^2 - 2$.

- **Reducible in $\mathbb{R}[x]$:** $f(x) = (x - \sqrt{2})(x + \sqrt{2})$, and neither $x + \sqrt{2}$ nor $x - \sqrt{2}$ is a unit. (Units in $\mathbb{R}[x]$ are the nonzero constants $\mathbb{R}^\times$.)
- **Irreducible in $\mathbb{Q}[x]$ and in $\mathbb{Z}[x]$,** as $\sqrt{2} \notin \mathbb{Q}$.

Theorem 2.1: Remainder Theorem (from lecture; cf. Lee, Thm. 11.1)

Let K be a field and $f(x) \in K[x]$. Take any $a \in K$. Then there is a polynomial $q(x) \in K[x]$ such that

$$f(x) = q(x)(x - a) + f(a).$$

Proof. By the division algorithm in $K[x]$, there are polynomials $q(x), r(x) \in K[x]$ such that

$$f(x) = q(x)(x - a) + r(x), \quad \text{where } r(x) = 0 \text{ or } \deg(r(x)) < 1.$$

So $r(x) = c$ for some constant $c \in K$.

Evaluating both sides at a :

$$f(a) = q(a)(a - a) + c = c.$$

Hence $f(x) = q(x)(x - a) + f(a)$. □

Definition 2.1: Root of a polynomial (from lecture; cf. Lee, Def. 11.1)

Let K be a field and $f(x) \in K[x]$. If $a \in K$, we say a is a *root* of $f(x)$ if $f(a) = 0$.

Theorem 2.2: Factor Theorem (from lecture; cf. Lee, Thm. 11.2)

Let K be a field and $f(x) \in K[x]$. Take any $a \in K$. Then a is a root of $f(x)$ if and only if $(x - a) \mid f(x)$.

Proof. ($\Rightarrow$) Suppose a is a root, so $f(a) = 0$. By the Remainder Theorem,

$$f(x) = q(x)(x - a) + f(a) = q(x)(x - a),$$

so $(x - a) \mid f(x)$.

($\Leftarrow$) Suppose $(x - a) \mid f(x)$. Then $f(x) = (x - a)g(x)$ for some $g(x) \in K[x]$. So $f(a) = (a - a)g(a) = 0$, and a is a root. □

Lemma 2.1: Root implies reducible (degree > 1) (from lecture; cf. Lee, Cor. 11.1)

Let K be a field, $f(x) \in K[x]$. If $\deg(f(x)) > 1$ and $f(x)$ has a root in K , then $f(x)$ is reducible.

Proof. Let $a \in K$ be a root. By the Factor Theorem,

$$f(x) = (x - a)g(x) \quad \text{for some } g(x) \in K[x].$$

Since $\deg(f(x)) > 1$, the factor $g(x)$ is not a constant, hence not a unit. So $f(x)$ is reducible. $\square$

Remark 2.1: The converse is false in general (from lecture)

$f(x) = (x^2 + 1)^2 \in \mathbb{R}[x]$ is reducible but has *no* real roots.

However, a degree-2 polynomial in $K[x]$ *is* reducible if and only if it has a root in K .

E.g. $f(x) = x^2 - 2$ is irreducible in $\mathbb{Q}[x]$ because $\sqrt{2}$ is irrational.

Checklist of Skills

- ☐ Prove uniqueness of factorisation in a PID, completing the proof that every PID is a UFD.
- ☐ Justify the cancellation argument: divisors of a unit are units.
- ☐ State (and explain) the fact that in any UFD, irreducible $\iff$ prime.
- ☐ Quote the result: R a UFD $\Rightarrow R[x]$ a UFD; identify the fraction field as the proof's main tool.
- ☐ State and prove the Remainder Theorem in $K[x]$.
- ☐ Define a root of $f(x) \in K[x]$.
- ☐ State and prove the Factor Theorem.
- ☐ Prove: if $\deg(f) > 1$ and f has a root in K , then f is reducible.
- ☐ Give an example showing the converse fails (e.g. $(x^2 + 1)^2$ in $\mathbb{R}[x]$).